NOTRE DAME OF MIDSAYAP COLLEGE
COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING
Midsayap, Cotabato

Proposed Research Agenda for CITE

PROGRAM	AGENDA	REGIONAL/LOCAL	NATIONAL	SUSTAINABLE DEVELOPMENT GOALS
Electronics Engineering Computer Engineering	1. **Local Language Processing and Machine Translation:** Research into natural language processing (NLP) tools that can handle Philippine languages and dialects. This includes developing machine translation systems, speech recognition, and text-to-speech systems for languages like Filipino, Cebuano, Ilocano, etc.	This aligns with goals for **promoting digital inclusion** and **enhancing regional communication**. It supports **local language technology** development and **cultural preservation** through **machine translation**.	This research fosters inclusivity by enabling more accessible communication, especially in rural areas, contributing to a "***Matatag***" (strong) and ***"Maginhawa***" (comfortable) life.	**SDG 4: Quality Education** - Promotes inclusive education by making content accessible in different languages. **SDG 10: Reduced Inequalities** - Enhances access to information and services, especially in marginalized communities.
	2. Cybersecurity Solutions for Small and Medium Enterprises (SMEs): Developing affordable and effective cybersecurity solutions tailored for SMEs in the Philippines to protect against increasing cyber threats and attacks.	This supports **enhancing regional cybersecurity** and protecting **SMEs** from **cyber threats**. It aligns with goals related to **securing digital infrastructure** and promoting **safe business operations**.	Supports inclusive and sustainable economic growth, ensuring that businesses are secure and resilient, contributing to a ***"Panatag"*** (secure) life.	**SDG 8: Decent Work and Economic Growth** - Strengthens the security of SMEs, promoting sustainable economic growth. **SDG 9: Industry, Innovation, and**

				Infrastructure - Focuses on innovation and infrastructure for business security and resilience.
	3. Disaster Management and Resilience Technologies: Creating advanced systems for disaster prediction, management, and recovery that leverage real-time data, AI, and IoT. This includes tools for early warning systems for typhoons, earthquakes, and floods.	This aligns with **disaster preparedness** and **resilience building** in the region. It supports goals for **improving disaster response** and **enhancing community resilience** through **advanced technologies**.	It ensures a secure and resilient environment, promoting preparedness and community resilience, crucial to achieving a ***"Panatag"*** society.	**SDG 11: Sustainable Cities and Communities** - Enhances urban resilience and disaster management. **SDG 13: Climate Action** - Promotes technologies for disaster resilience in response to climate-related hazards.
	4. E-Government and Public Service Optimization: Research ways to improve e-government platforms and public service delivery through digital means, including developing secure and efficient public administration and citizen engagement systems.	These support **improving public services** and **government efficiency**. It aligns with goals related to **digital governance**, **service delivery improvement**, and **enhancing citizen engagement**.	Enhances access to efficient and reliable government services, contributing to a streamlined society where people enjoy a ***"Maginhawa"*** life due to better public service.	**SDG 16: Peace, Justice, and Strong Institutions** - Supports transparent, efficient public services and governance. **SDG 9: Industry, Innovation, and Infrastructure** - Improves government service delivery through digital innovation.

	5. Agricultural Technology and Precision Farming: Developing computer-based solutions for enhancing agricultural productivity, such as innovative farming technologies, precision agriculture tools, and agricultural data analytics tailored to Philippine farming practices.	This aligns with **advancing agricultural practices** and **supporting rural development**. It supports goals related to **precision farming**, **agricultural innovation**, and **increasing food security** in the region.	Promotes sustainable agriculture, increasing productivity, and supporting food security, which is crucial for creating a ***"Matatag"*** and "***Maginhawa"*** life for Filipinos.	**SDG 2: Zero Hunger** - Enhances agricultural productivity and sustainability. **SDG 12: Responsible Consumption and Production** - Promotes sustainable farming practices and resource management.
	6. Smart City Solutions and Urban Planning: Research smart city technologies that can address urban challenges in Philippine cities, such as traffic management, waste management, and energy efficiency through IoT and data analytics.	This supports **smart city development** and **urban planning**. It aligns with goals related to **sustainable urban growth**, **improving urban infrastructure**, and **enhancing quality of life** in urban areas.	It focuses on improving urban living through smart infrastructure, enabling a better quality of life and convenience, and is aligned with a ***"Maginhawa"*** and "***Panatag***" future.	**SDG 11: Sustainable Cities and Communities** - Focuses on sustainable urban development and smart city technologies. **SDG 9: Industry, Innovation, and Infrastructure** - Encourages urban planning and infrastructure innovation.
	7. Renewable Energy and Environmental Monitoring: Exploring computer-based systems for monitoring and managing renewable energy resources and environmental impact assessment	This aligns with **sustainable energy solutions** and **environmental protection** goals. It supports **renewable energy adoption**, **environmental monitoring**,	Contributes to sustainability goals by promoting clean energy solutions and environmental monitoring, ensuring a	**SDG 7: Affordable and Clean Energy** - Promotes using renewable energy technologies.

	tools to support sustainable development.	and **reducing carbon footprint** in the region.	"***Panatag***" life through ecological preservation.	**SDG 13: Climate Action** - Supports environmental monitoring and actions to mitigate climate change.
	8. Digital Inclusion and Accessibility: Research ways to bridge the digital divide by developing affordable and accessible technologies for underserved communities, including low-cost computing devices, internet access solutions, and digital literacy programs.	This supports goals related to **inclusive technology access** and **reducing the digital divide**. It aligns with enhancing **accessibility** for **all individuals**, including those with disabilities, and promoting **equitable digital opportunities**.	Supports the goal of reducing inequality by providing equitable access to digital services, ensuring a "***Matatag***" and "***Maginhawa***" society where everyone has opportunities to grow.	**SDG 10: Reduced Inequalities** - Ensures digital access for marginalized populations. **SDG 9: Industry, Innovation, and Infrastructure** - Encourages equitable access to digital technologies.
	9. Communication Systems for Remote Areas: Developing innovative communication technologies that provide reliable connectivity in remote or isolated regions, including satellite communication, mesh networks, and low-power wireless solutions.	This aligns with **improving connectivity** in **remote areas**. It supports goals for **expanding communication infrastructure** and **enhancing access to digital services** in underserved regions.	Bridges the digital divide, ensuring all Filipinos, especially in remote areas, have access to essential communication services, supporting a "***Panatag***" and "***Matatag***" life.	**SDG 9: Industry, Innovation, and Infrastructure** - Enhances connectivity and communication infrastructure in remote areas. **SDG 10: Reduced Inequalities** - Bridges the digital divide by improving access in underserved regions.

	10. Microelectronics and VLSI Design: Investigating new semiconductor materials and technologies to improve electronic devices and systems' performance, efficiency, and cost-effectiveness.	This supports **advancing microelectronics** and **integrated circuit design**. It aligns with goals related to **technological innovation**, **supporting the local electronics industry**, and contributing to **regional technological development**.	Drives innovation and industrial growth, positioning the country as a leader in advanced technologies, contributing to economic competitiveness and a "***Maginhawa***" life for Filipinos.	**SDG 9: Industry, Innovation, and Infrastructure** - Advances technological innovation in microelectronics. **SDG 8: Decent Work and Economic Growth** - Drives economic growth through advanced electronics and technological development.
COMPUTER SCIENCE (CS) PROGRAM	**1. Emerging Technologies in Computer Science:** These technologies often promise to solve complex problems, enhance capabilities, or create new opportunities. They typically include advancements in areas such as artificial intelligence (AI), machine learning, quantum computing, blockchain, augmented reality (AR), virtual reality (VR), and the Internet of Things (IoT).	This research can support **regional innovation** and **technology adoption**, which may be a focus in the plan's goals for promoting **digital transformation** and enhancing **technological infrastructure**. It aligns with fostering a **competitive technology sector** in the region.	Fosters innovation and technological leadership, helping Filipinos become globally competitive and contributing to a prosperous "***Maginhawa***" future.	**SDG 9: Industry, Innovation, and Infrastructure** - Encourages innovation and the adoption of advanced technologies to strengthen infrastructure and industry. **SDG 8: Decent Work and Economic Growth** - Promotes technological advancement to drive sustainable economic growth.

	2. Big Data Analytics for Predictive Modeling: Research techniques for analyzing large datasets to build predictive models, examining applications in healthcare, finance, or environmental monitoring.	This aligns with regional priorities related to **economic development** and **smart governance**. Big data analytics can improve **decision-making processes** in various sectors, such as agriculture, health, and urban planning, supporting the region's **growth and development goals**.	Enhances decision-making across industries, supporting inclusive economic growth and innovation, leading to a "***Maginhawa***" and "***Panatag***" life where Filipinos benefit from informed policies and business strategies.	**SDG 9: Industry, Innovation, and Infrastructure** - Enhances industry decision-making using data-driven insights. **SDG 11: Sustainable Cities and Communities** - Contributes to urban planning and smart city solutions through data analytics. **SDG 13: Climate Action** - Utilizes big data for climate-related predictive modeling and disaster prevention.
	3. Ethical Implications of Artificial Intelligence: Study the ethical considerations surrounding AI technologies, including bias, privacy, and the societal impact of automation and decision-making systems.	Research on AI ethics is crucial for creating **policies and frameworks** that ensure **responsible technology use**. This aligns with goals related to promoting **ethical practices** and safeguarding **citizens' rights and privacy** in the technological advancement landscape.	Ensures responsible technological growth, safeguards Filipinos' rights, promotes societal fairness, and aligns with a secure and just "***Panatag***" life.	**SDG 16: Peace, Justice, and Strong Institutions** - Addresses the ethical use of AI in governance, justice, and societal systems. **SDG 10: Reduced Inequalities** - Ensures AI is developed and

				implemented in ways that reduce social inequalities.
	4. Cybersecurity Threat Detection and Prevention: Research new approaches for detecting and preventing cybersecurity threats, including developing intrusion detection systems and real-time threat analysis tools.	This supports goals related to **securing digital infrastructure** and protecting **data integrity**. It is aligned with enhancing **regional cybersecurity measures** and promoting a **safe digital environment** for businesses and individuals.	It protects individuals and institutions from cyber threats, ensuring secure digital transactions and data privacy and contributing to a "***Panatag***" life with enhanced public and business safety.	**SDG 9: Industry, Innovation, and Infrastructure** - Strengthens infrastructure by protecting against cyber threats. **SDG 16: Peace, Justice, and Strong Institutions** - Supports the integrity and security of institutions through effective cybersecurity measures.
	5. Design and Evaluation of User Interfaces for Accessibility: Investigate the design of user interfaces that improve accessibility for individuals with disabilities, focusing on usability testing and implementing inclusive design principles.	This aligns with goals for **inclusive development** and improving **access to technology** for all individuals, including those with disabilities. It supports the creation of **equitable digital experiences** and advances in **accessibility standards** in the region.	Promotes digital inclusion by ensuring technology is accessible to all, including marginalized sectors, supporting a "***Maginhawa***" and "***Matatag***" society where everyone can participate in digital innovations.	**SDG 10: Reduced Inequalities** - Promotes digital inclusion by designing accessible user interfaces. **SDG 4: Quality Education** - Enhances access to education platforms and digital resources through accessible design.

	6. Development of Educational Games for Learning Programming and ITE Concepts: Create and evaluate educational games designed to teach fundamental programming concepts to make learning more engaging and effective for beginners.	This supports **educational goals** by enhancing **STEM education** and increasing **digital literacy**. It aligns with the plan's focus on developing **skills and competencies** in the workforce to foster innovation and **technological capability**.	Enhancing educational opportunities improves education through engaging learning tools, supporting skill development and lifelong learning, and contributing to a "***Maginhawa***" life.	**SDG 4: Quality Education** - Improves education outcomes by using interactive games to teach programming, especially for younger learners. **SDG 8: Decent Work and Economic Growth** - Encourages skill development in programming, contributing to future employment opportunities in tech.
INFORMATION TECHNOLOGY (IT) PROGRAM	**1. Software Development:** Research and conduct of designing, creating, testing, and maintaining software applications and systems. It involves a systematic approach to building software that meets users' or businesses' needs and requirements.	This aligns with **regional innovation** and the development of **new technologies**. It supports the **digital transformation** and **technology adoption** goals by creating **software solutions** to drive **economic growth** and **improve regional services**.	Drives innovation and technological growth, enabling businesses and individuals to adopt modern solutions, contributing to a "***Maginhawa***" (comfortable) life where digital solutions improve daily activities and services.	**SDG 9: Industry, Innovation, and Infrastructure** - Advances innovation by developing software solutions for various industries. **SDG 8: Decent Work and Economic Growth** - Encourages technological development, improving productivity, and

				supporting economic growth.
	2. Multimedia Systems: Development integration and management of multiple forms of media—such as text, audio, video, graphics, and animation—within computing systems to create, process, and deliver interactive and engaging content. These systems are designed to handle the diverse requirements of multimedia content, including its creation, storage, manipulation, and presentation. This agendum may include Game Development, e-learning Systems, Interactive Systems, and Information Kiosks.	Research in multimedia systems supports **creative industries** and **digital media development**. It aligns with the plan's goals to promote **cultural and creative sectors** and enhance **regional communication** through innovative **multimedia applications**.	Enhances digital communication and education through multimedia technologies, supporting a "***Maginhawa***" and "***Matatag***" (strong) society where people have access to modern, interactive platforms for learning and engagement.	**SDG 4: Quality Education** - Supports educational tools through multimedia content, enhancing digital learning experiences. **SDG 10: Reduced Inequalities** - Facilitates access to information through multimedia platforms, bridging gaps for underserved communities.
	3. Network Design and Implementation: Focuses on exploring and advancing methodologies, technologies, and best practices for creating and deploying network infrastructures. This research addresses challenges and opportunities in designing and implementing efficient, scalable, secure, and adaptable networks for evolving technological needs.	This aligns with **improving digital infrastructure** and supporting **connectivity goals** in the region. It is relevant for **enhancing network capabilities**, supporting **smart cities**, and improving **access to digital services**.	Improves communication infrastructure, ensuring that Filipinos, especially in remote and underserved areas, can access reliable internet services, supporting a "***Panatag***" (secure) and "***Matatag***" society through digital connectivity.	**SDG 9: Industry, Innovation, and Infrastructure** - Strengthens communication infrastructure by designing reliable and secure networks. **SDG 10: Reduced Inequalities** - Ensures equal access to internet services through improved network infrastructure in

				remote and underserved areas.
	4. Server Farm Configuration and Management: Involves investigating and advancing methodologies, technologies, and best practices for the setup, operation, and optimization of server farms—large-scale collections of servers used to deliver applications, services, and data. This research area focuses on improving server farms' efficiency, scalability, security, and reliability to meet the demands of modern IT environments.	This supports **technological infrastructure** development and **data management** capabilities. It aligns with goals related to **efficient IT resource management**, **supporting regional businesses**, and **ensuring reliable IT services**.	Promotes efficient and sustainable IT infrastructure management, contributing to a "***Maginhawa***" life by supporting the country's growing digital economy with reliable, scalable IT systems.	**SDG 7: Affordable and Clean Energy** - Optimizes server farms for energy efficiency, supporting clean and sustainable energy use in IT. **SDG 9: Industry, Innovation, and Infrastructure** - Supports innovation and resilience in data management through efficient server management.
	5. **IT Management**: Focuses on exploring and advancing the strategies, practices, and technologies involved in effectively overseeing and directing information technology (IT) resources and operations within an organization. This research area aims to optimize the use of IT to support organizational goals, improve efficiency, and drive innovation.	Research in IT management aligns with **effective governance** and **strategic IT planning**. It supports **improving organizational efficiency**, **enhancing decision-making**, and **driving innovation** in regional businesses and institutions.	Strengthens businesses' and institutions' management and operational efficiency through effective IT solutions, enabling sustained economic growth and contributing to a "***Maginhawa***" and "***Panatag***" life for all.	**SDG 8: Decent Work and Economic Growth** - Enhances organizational efficiency, promoting sustainable economic growth through IT management. **SDG 9: Industry, Innovation, and Infrastructure** - Improves industry processes and operations through

				effective IT systems management.
INFORMATION SYSTEMS (IS) PROGRAM	**1. Software Development:** It involves exploring and advancing the methodologies, tools, techniques, and best practices associated with the creation, deployment, and maintenance of software applications and systems within the broader context of information systems. This research agenda focuses on improving software development processes' effectiveness, efficiency, and quality to support organizational needs better and address emerging business process challenges.	This aligns with **regional innovation** and **technology development** goals. It supports **digital transformation** and the creation of **software solutions** that drive **economic growth** and **enhance regional services**.	Enhances economic productivity and innovation by providing businesses and individuals with cutting-edge software solutions, contributing to a "***Maginhawa***" (comfortable) life where technology simplifies processes and services.	**SDG 9: Industry, Innovation, and Infrastructure** - Advances innovation by creating software solutions to support industry and infrastructure. **SDG 8: Decent Work and Economic Growth** - Enhances productivity and fosters economic growth by developing practical software tools.
	2. Information Systems Planning: It involves investigating and advancing the strategies, methodologies, and best practices for effectively planning and managing organization information systems. This research focuses on aligning information systems with organizational goals, optimizing resource allocation, and ensuring technology investments deliver maximum value.	This supports **strategic IT planning** and **effective governance**. It aligns with the goals of **improving organizational efficiency**, **enhancing decision-making**, and **supporting regional development** through well-planned **information systems**.	Ensures the strategic use of information systems to support institutional growth and development, fostering a "***Panatag***" (secure) life by strengthening organizational operations and decision-making processes.	**SDG 9: Industry, Innovation, and Infrastructure** - Supports strategic infrastructure planning by optimizing information systems. **SDG 16: Peace, Justice, and Strong Institutions** - Strengthens institutional capacity through well-planned, reliable information systems

				for governance and decision-making.
	3. Analysis and Design of a Sufficiently Complex Business System: It focuses on developing advanced methodologies, tools, and frameworks for systematically studying and creating intricate business systems. These complex systems involve numerous interconnected components, processes, and stakeholders. The research agenda aims to enhance such systems' effectiveness, efficiency, and adaptability to meet dynamic business needs and technological challenges.	This aligns with **developing complex business solutions** and supporting **business process improvement**. It is relevant for goals related to **enhancing business operations**, **fostering innovation**, and **supporting regional enterprises**.	Supports economic growth by optimizing business operations through well-designed systems, contributing to a "***Maginhawa***" and "***Matatag***" (strong) society where businesses are competitive and efficient.	**SDG 8: Decent Work and Economic Growth** - Improves business operations and drives economic growth by designing efficient, scalable business systems. **SDG 9: Industry, Innovation, and Infrastructure** - Encourages innovation in business systems, supporting sustainable industry infrastructure.
	4. Data Analytics and Business Intelligence: This research agenda focuses on advancing the fields of Data Analytics and Business Intelligence (BI) by exploring innovative techniques, tools, and methodologies to enhance data-driven decision-making in organizations.	This supports **data-driven decision-making** and **improving business insights**. It aligns with goals related to **economic development**, **smart governance**, and **enhancing regional competitiveness** through **advanced data analytics** and **business intelligence**.	Drives informed decision-making and business innovation by leveraging data analytics, contributing to a "***Maginhawa***" and "***Panatag***" life through improved business strategies and economic competitiveness.	**SDG 9: Industry, Innovation, and Infrastructure** - Supports informed industry decision-making through data analytics and business intelligence. **SDG 8: Decent Work and Economic Growth** - Promotes sustainable economic growth by

				leveraging data insights to enhance business strategies and operations.